test1

Master's Degree Project Report

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1 Abstract

This section will be filled in at the end of the project and will contain a concise summary of the study, including:

- Background
- Objective
- Methods
- Key results
- Conclusion

2 Introduction

2.1 Background: Cardiomyopathy

Cardiomyopathy is a group of diseases that affect the heart muscle, leading to structural and functional abnormalities that impair the heart's ability to pump blood efficiently. These conditions can result in serious clinical outcomes, including heart failure, arrhythmias, and sudden cardiac death.

Many forms of cardiomyopathy have a genetic basis. Inherited mutations have been associated with several types of the disease, such as hypertrophic, dilated, restrictive, and arrhythmogenic cardiomyopathy, as well as left ventricular non-compaction and ion channelopathies. These mutations often affect genes involved in heart muscle structure and function, including those related to the sarcomere, cytoskeleton, and ion channels.

2.2 Aim and Research Questions

The aim of this project is to identify and evaluate genetic variants associated with cardiomyopathy using DNA sequencing data. By comparing the patient's DNA sequence to established cardiomyopathy-associated gene panels, potentially pathogenic variants can be detected and assessed. Using next-generation sequencing (NGS) technologies, this project seeks to identify genetic variants that may contribute to the development and progression of cardiomyopathy and to evaluate their relevance in a diagnostic context.

3 Materials and Methods

3.1 Dataset Description

3.1.1 Data Source

The dataset used in this project consists of DNA sequencing data obtained from patients diagnosed with cardiomyopathy. The data were provided as part of a master-level genomics course and were used for educational and research purposes.

3.1.2 Number of Samples

The dataset consists of multiple patient samples derived from cardiomyopathy cases. In this project, the analysis focuses on **Sample 1**, which is provided as paired-end sequencing data consisting of two read files (forward and reverse reads).

3.1.3 File Format

The sequencing data were provided in FASTQ format, containing the raw reads generated by next-generation sequencing and used as the input for the analysis.

3.2 Analysis Environment

- **Workflow platform:** Galaxy
- **Programming language:** R
- **Reporting framework:** R Markdown

Version control and collaboration were managed using **GitHub**, allowing transparent tracking of changes and shared development of the analysis. **Microsoft Teams** was used as the primary communication platform to coordinate the workflow, discuss results, and share progress with each other and with the course instructor.

4 Analysis Workflow

The overall analysis workflow is designed to identify and characterize genetic variants from DNA sequencing data in a systematic and reproducible manner. The workflow includes quality control of raw reads, read preprocessing, alignment to a reference genome, variant calling, and variant filtering and annotation. Each step of the analysis is briefly introduced here and discussed in detail in the following subsections.

4.1 Quality Control of Raw Reads

Quality control of raw sequencing reads was performed to evaluate data quality before downstream analysis. For this purpose, the tool **Falco** was used. A `.fastQ.gz` file was provided as input, and Falco generates several plots as output. Each plot highlights a specific aspect of the sequencing data, such as base quality scores, GC content, sequence length distribution, and the presence of adapter contamination, and provides a quality assessment in the form of **pass, warn, or fail**.

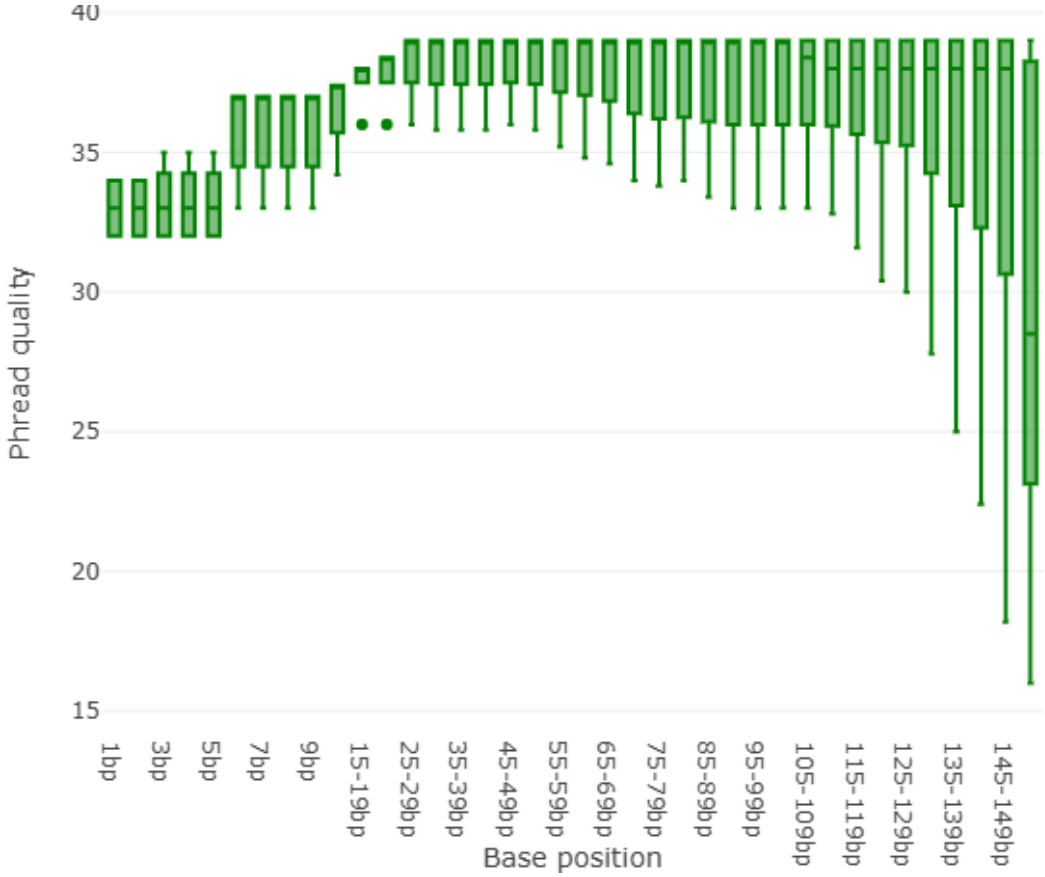
4.1.1 Quality Control Results ** here will add the results for r1 also!

The quality control results were examined to determine whether read trimming or filtering was required before alignment. Key observations from the quality reports are summarized and interpreted in this section.

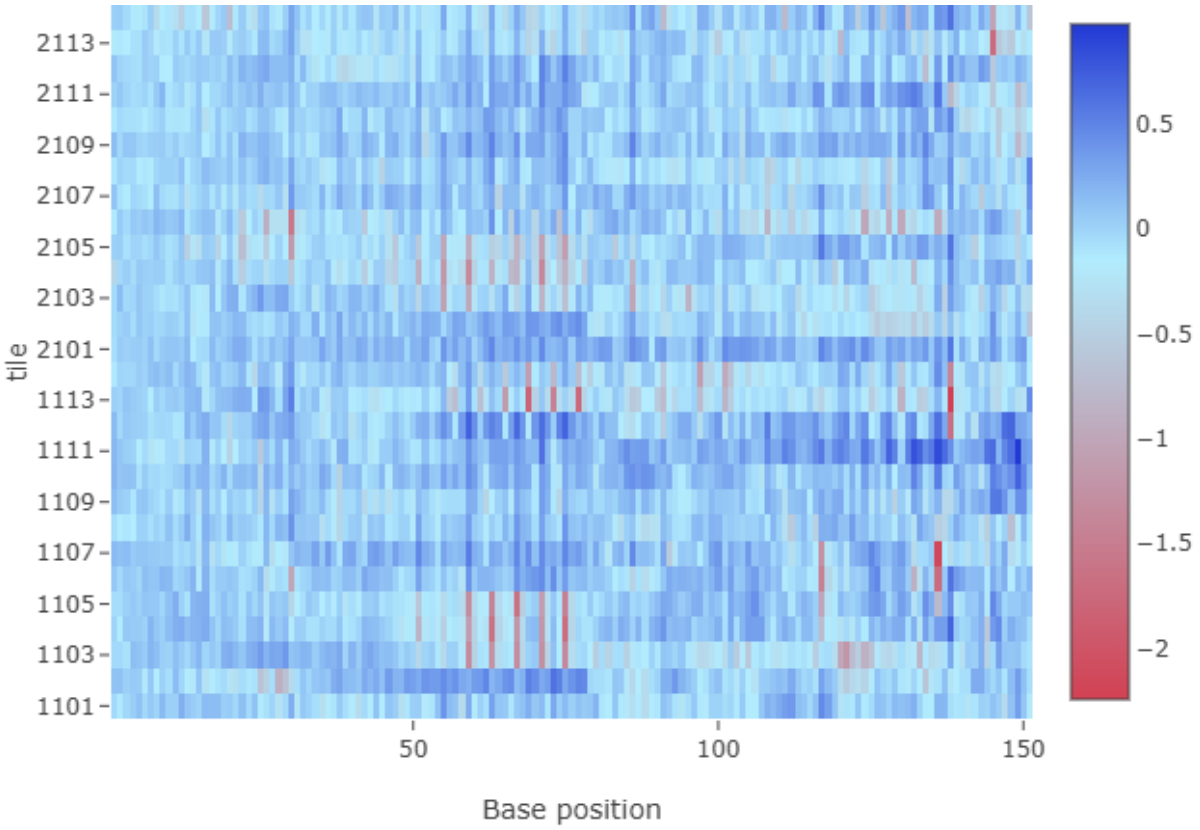
4.1.2 Basic Statistics

Measure	Value
Filename	Sample1_0394321_L001_R1_001_fastq_gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1424257
Sequences Flagged As Poor Quality	0
Sequence length	32 - 151
%GC:	43

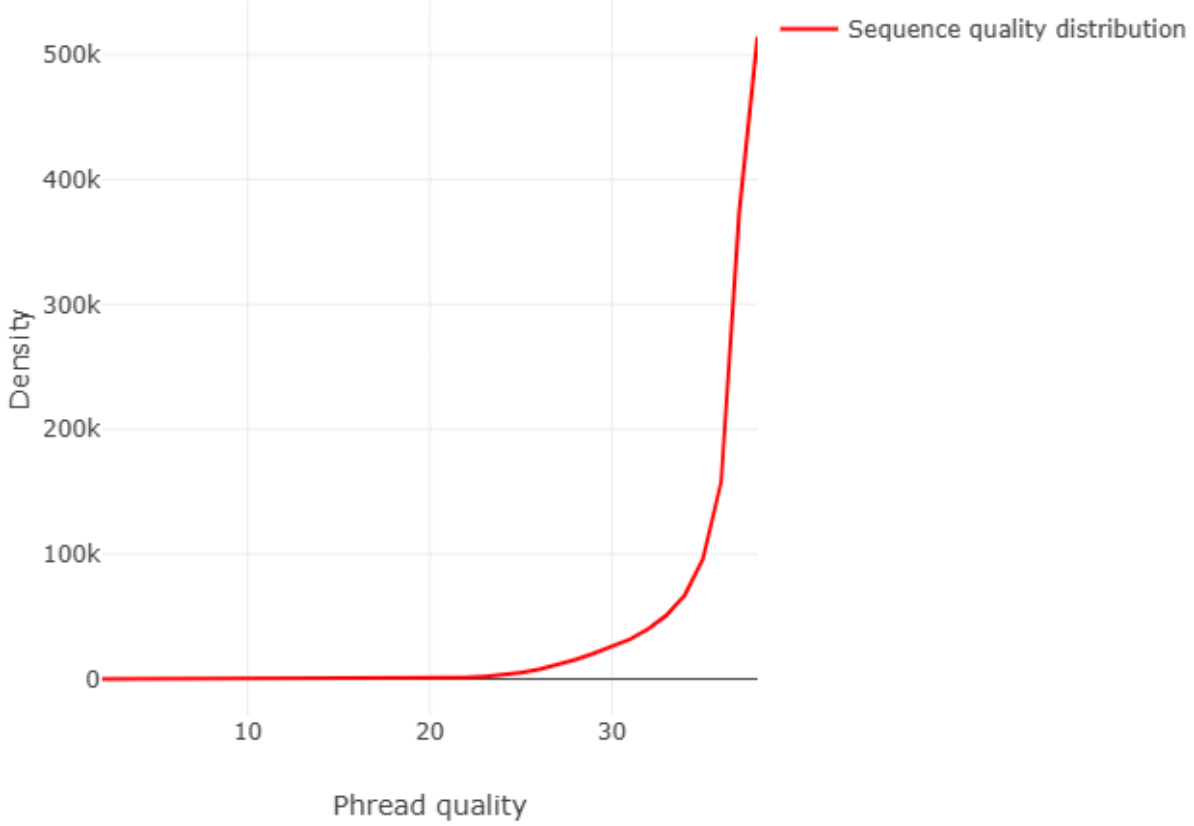
4.2 Per base sequence quality: pass



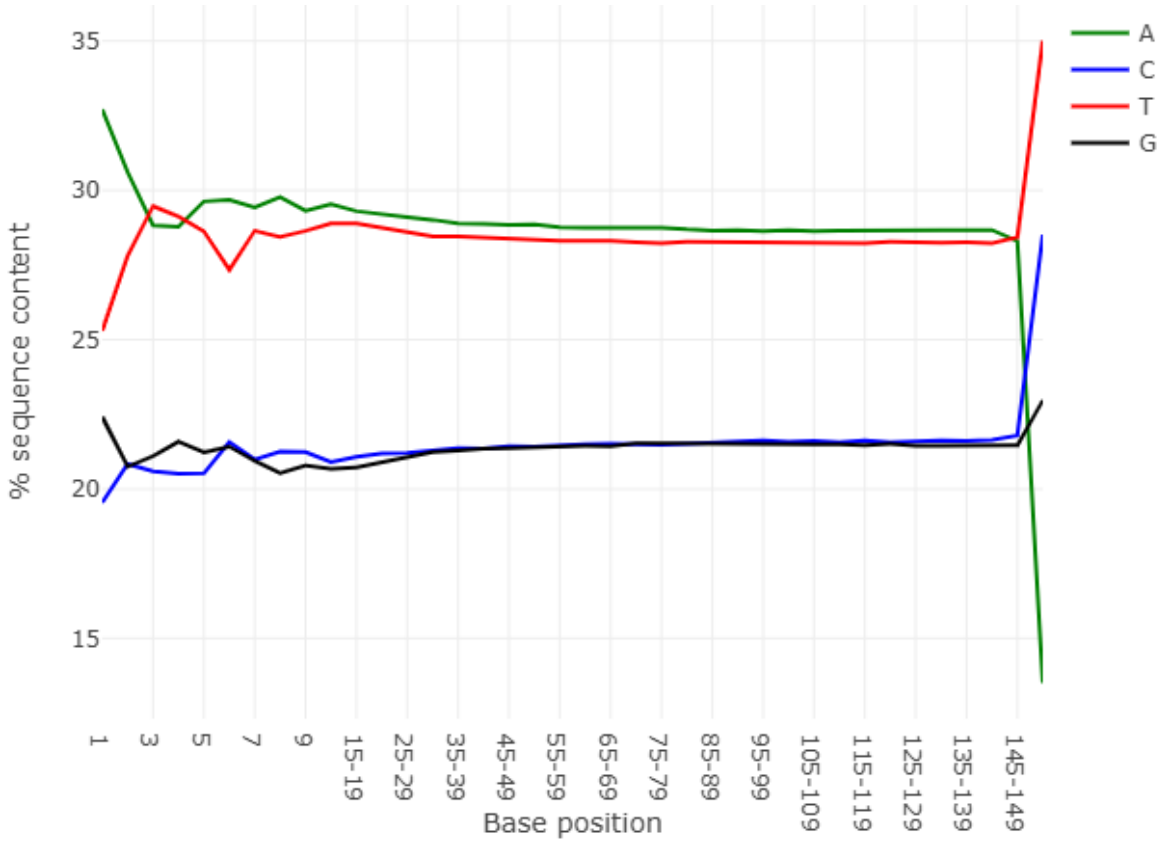
4.3 Per tile sequence quality : pass



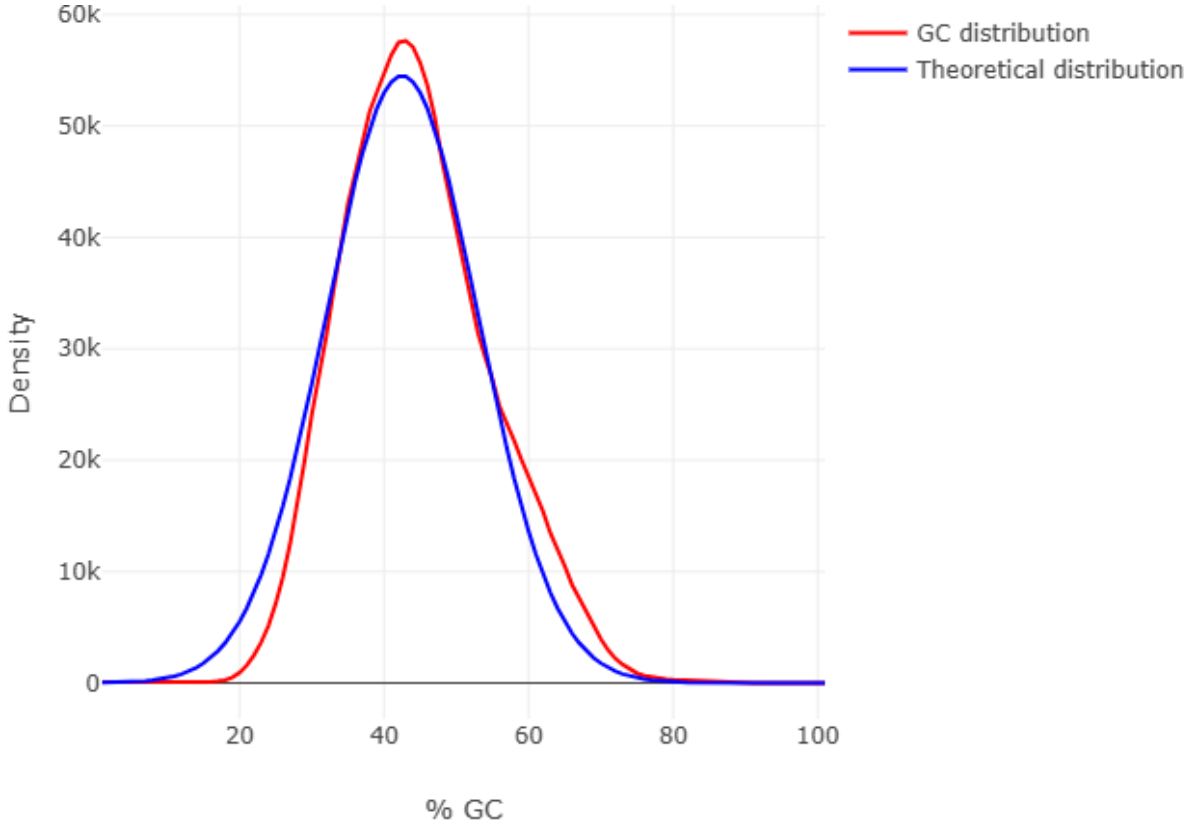
4.4 Per sequence quality scores : pass



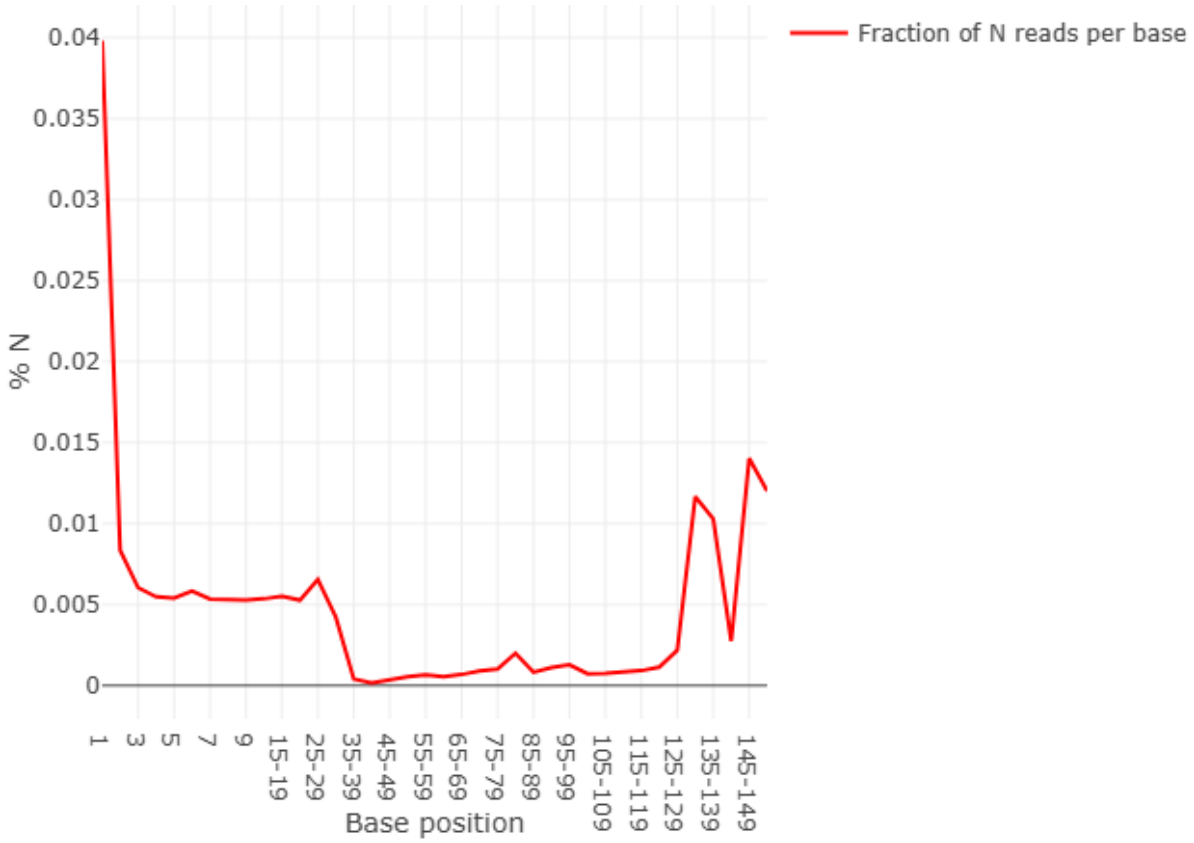
4.5 Per base sequence content : fail



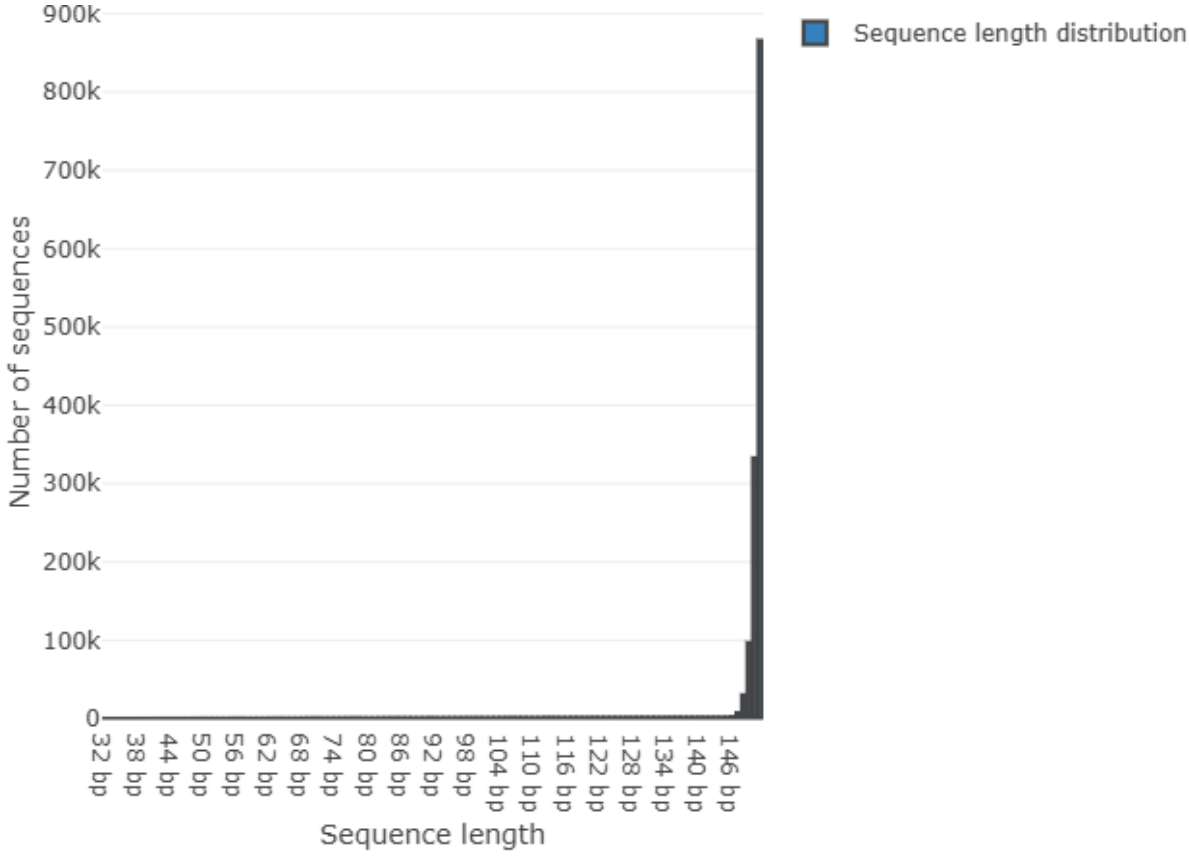
4.6 Per sequence GC content: pass



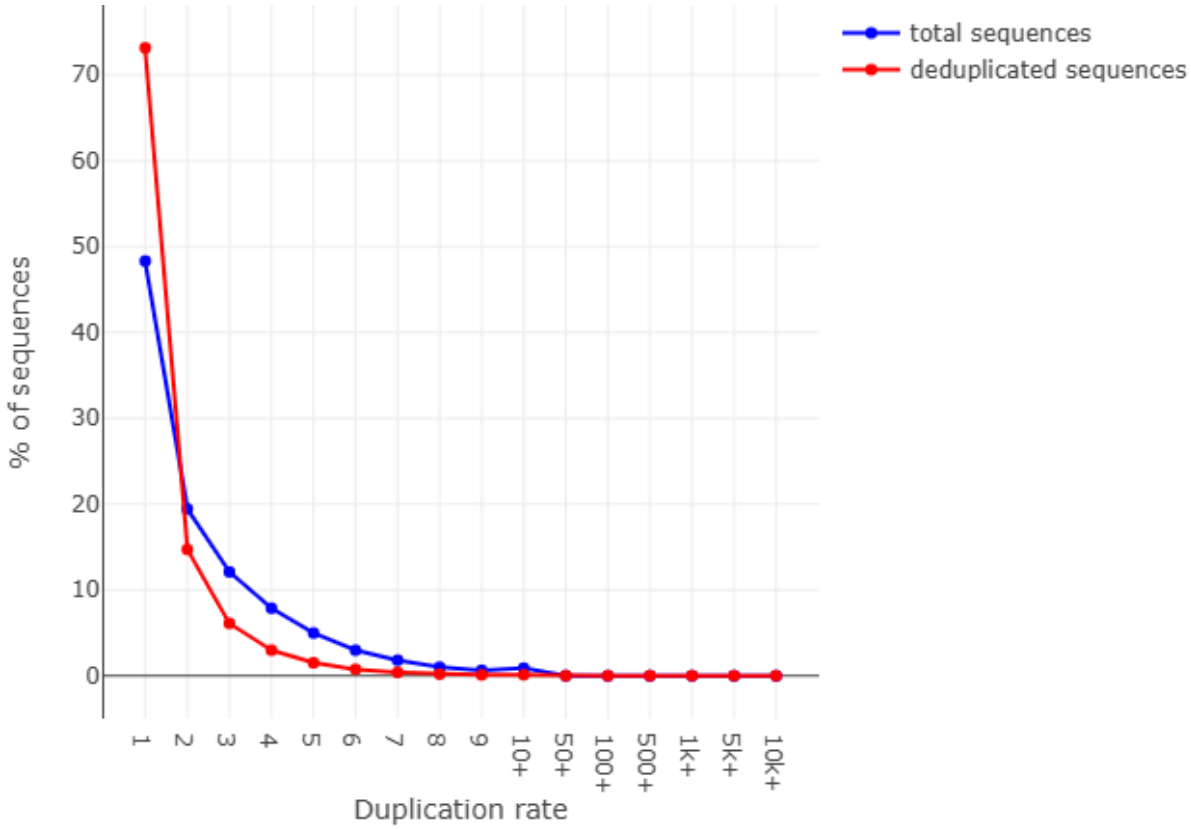
4.7 Per base N content : pass



4.8 Sequence Length Distribution : warn



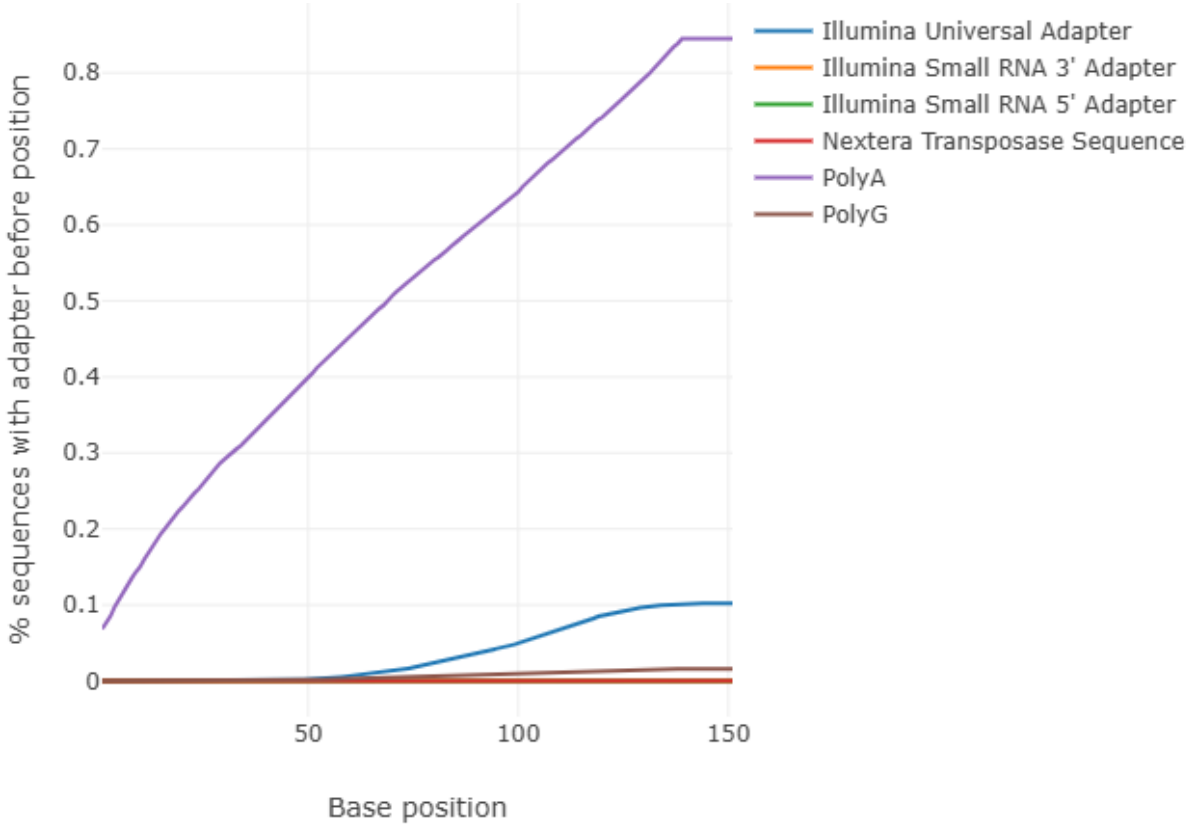
4.9 Sequence Duplication Levels : warn



4.10 Overrepresented sequences : pass

No overrepresented sequences

4.11 Adapter Content : pass



4.12 Read Trimming and Filtering ** see the jero doc!

Based on the initial quality assessment, reads were trimmed to remove low-quality bases and sequencing adapters, ensuring that only high-quality sequences were retained for downstream analysis.

Tool used: (`Trimmomatic`)

4.12.1 Trimming Parameters

Tool Parameters

Input Parameter	Value
Single-end or paired-end reads?	pair_of_files
Input FASTQ file (R1/first of pair)	
Input FASTQ file (R2/second of pair)	
Perform initial ILLUMINACLIP step?	true
Select standard adapter sequences or provide custom?	standard

Input Parameter	Value
Adapter sequences to use	TruSeq3 (paired-ended, for MiSeq and HiSeq)
Maximum mismatch count which will still allow a full match to be performed	2
How accurate the match between the two 'adapter ligated' reads must be for PE palindrome read alignment	30
How accurate the match between any adapter etc. sequence must be against a read	10
Minimum length of adapter that needs to be detected (PE specific/palindrome mode)	8
Always keep both reads (PE specific/palindrome mode)?	true
Select Trimmomatic operation to perform	SLIDINGWINDOW
Number of bases to average across	4
Average quality required	20
Select Trimmomatic operation to perform	CROP
Number of bases to keep from the start of the read	145
Select Trimmomatic operation to perform	HEADCROP
Number of bases to remove from the start of the read	7
Select Trimmomatic operation to perform	MINLEN
Minimum length of reads to be kept	50
Quality score encoding	Nothing selected.
Output trimlog file?	false
Output trimmomatic log messages?	false

4.12.2 Quality Control Results after trimming

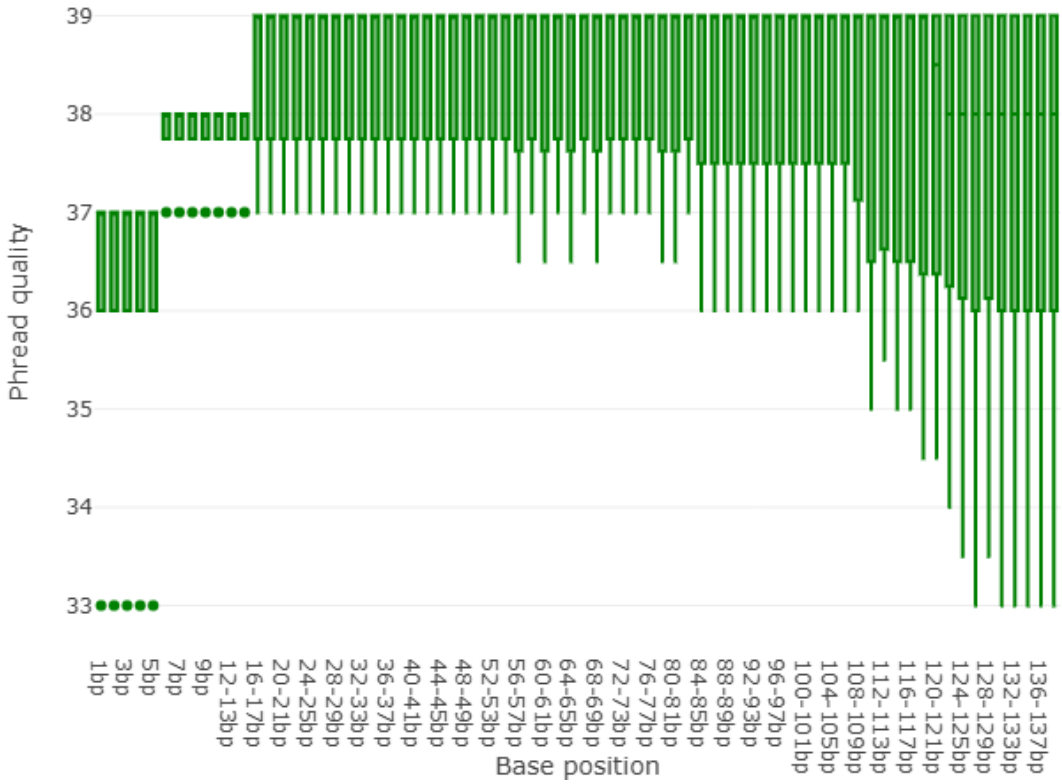
4.12.2.1 - Sample1_R1_fastq_gz_R1 paired

4.13 Basic Statistics: pass

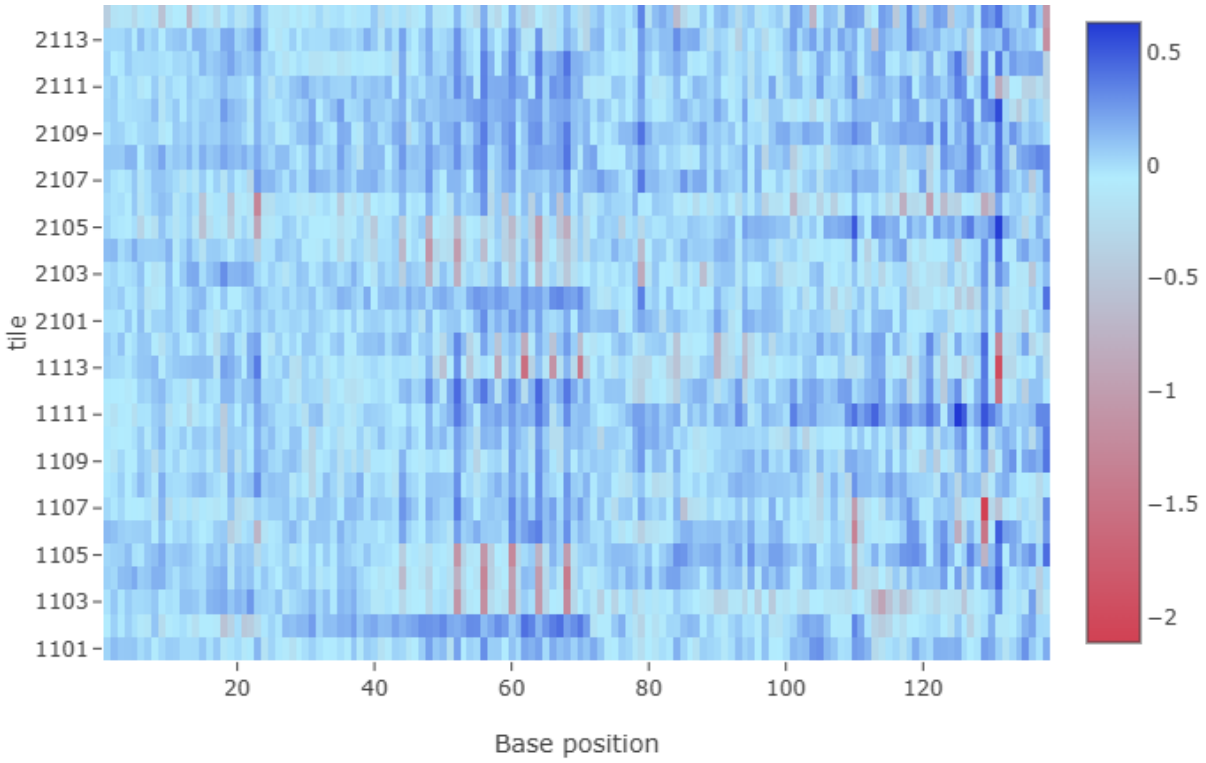
Measure	Value
Filename	Trimmomatic on Sample1_0394321_L001_R1_001_fastq_gz_R1 paired_
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1206159

Measure	Value
Sequences Flagged As Poor Quality	0
Sequence length	50 - 138
%GC:	42

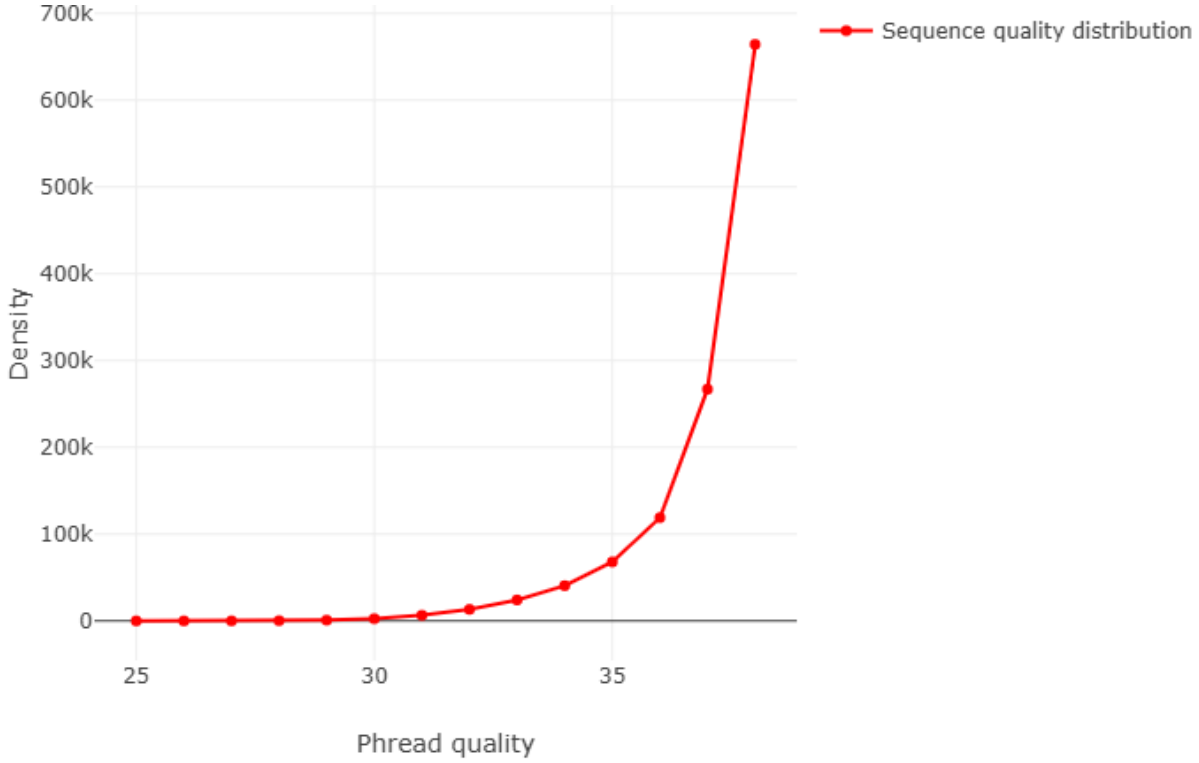
4.14 Per base sequence quality: pass



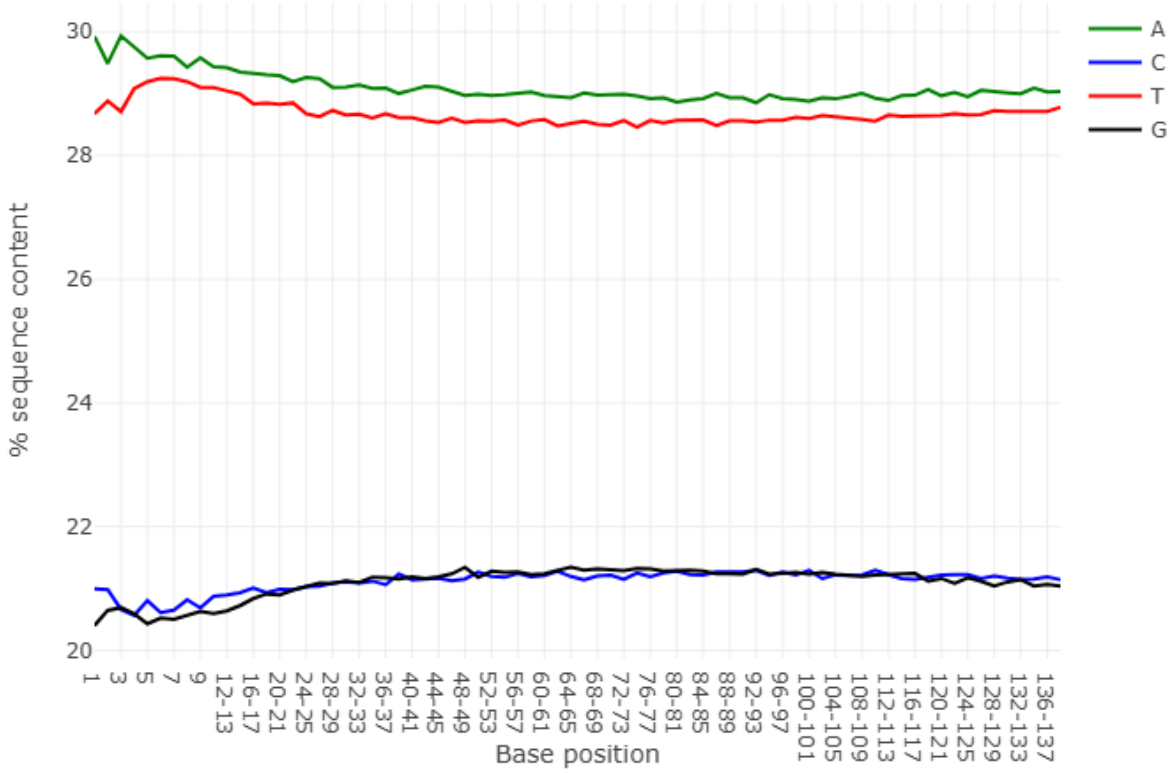
4.15 Per tile sequence quality: pass



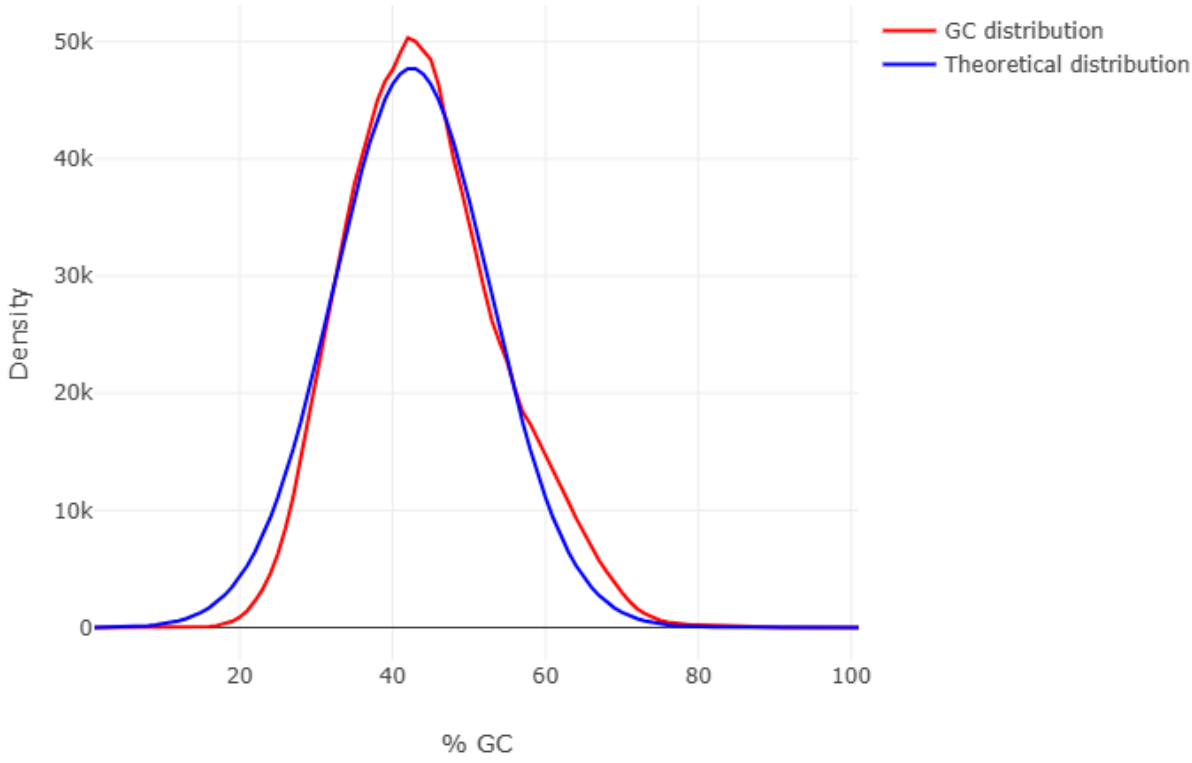
4.16 Per sequence quality scores: pass



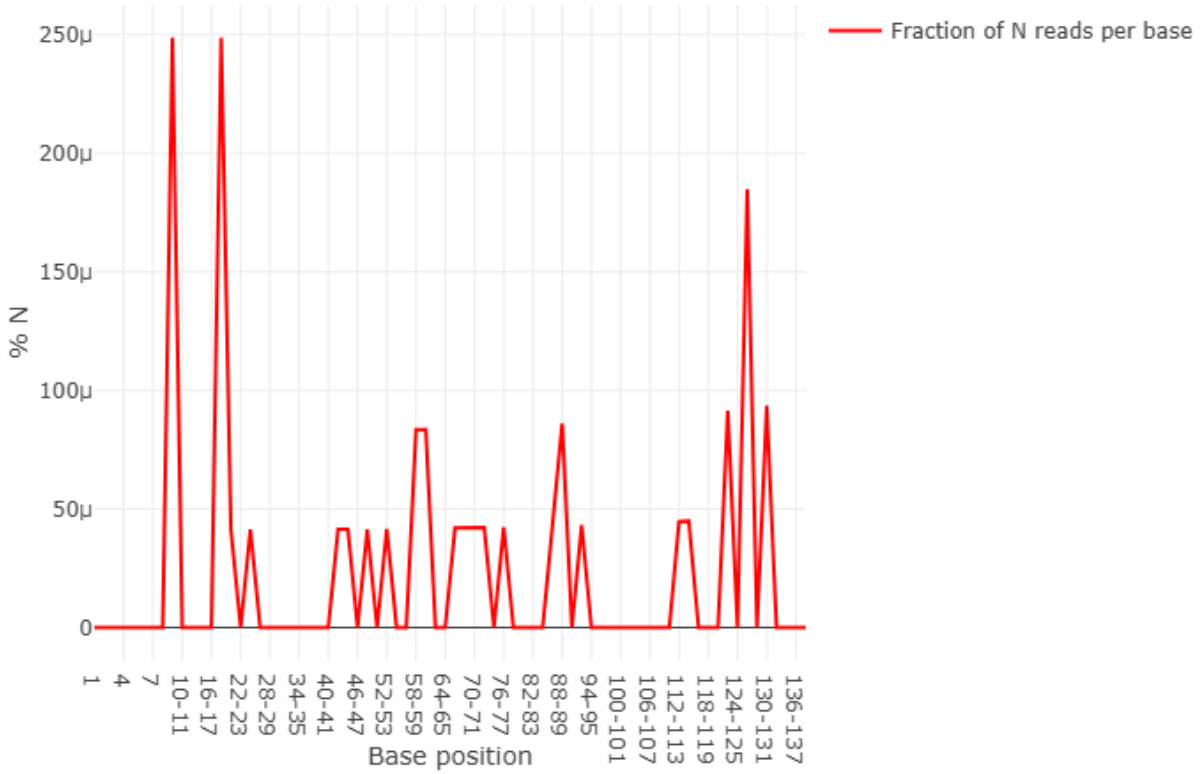
4.17 Per base sequence content : pass



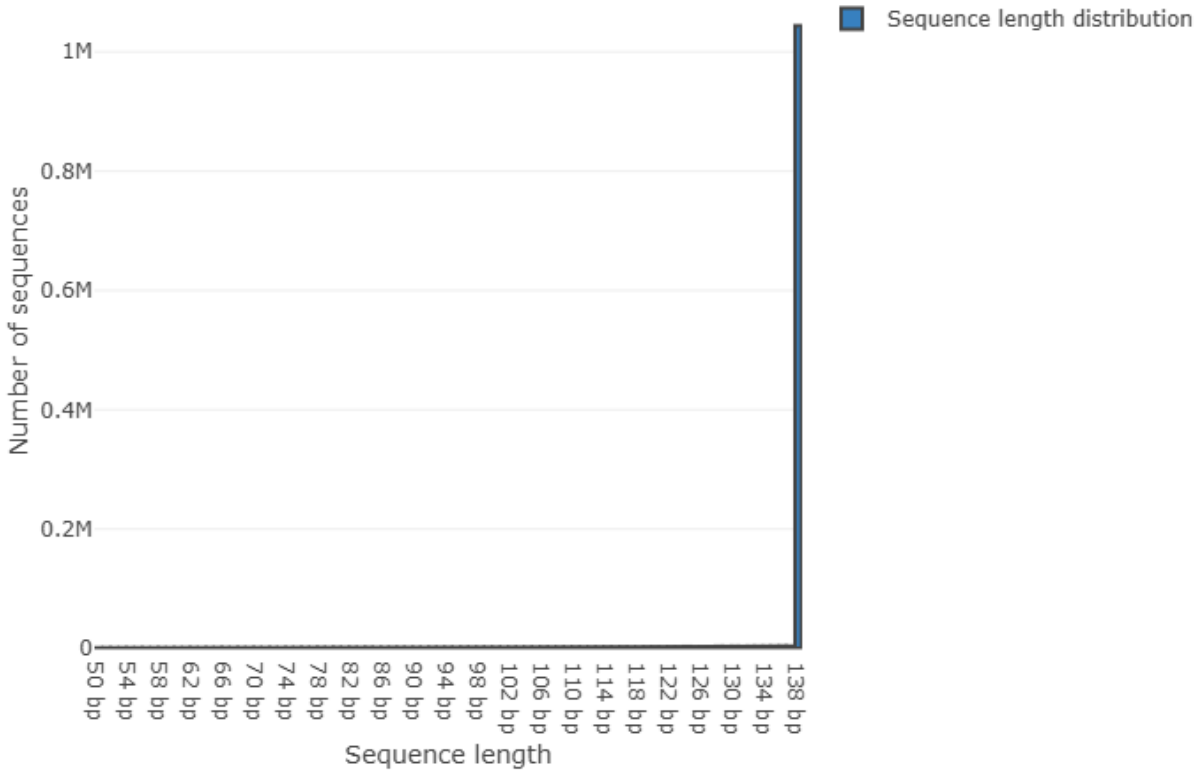
4.18 Per sequence GC content: pass



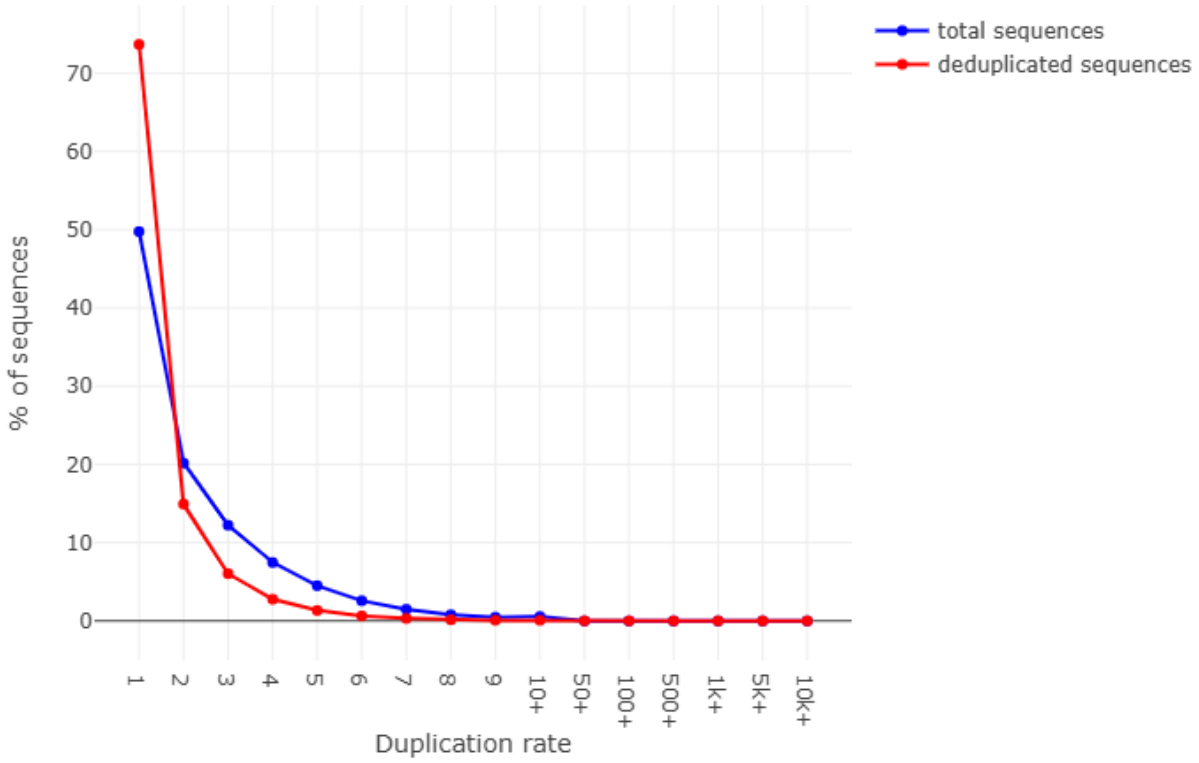
4.19 Per base N content : pass



4.20 Sequence Length Distribution : warn



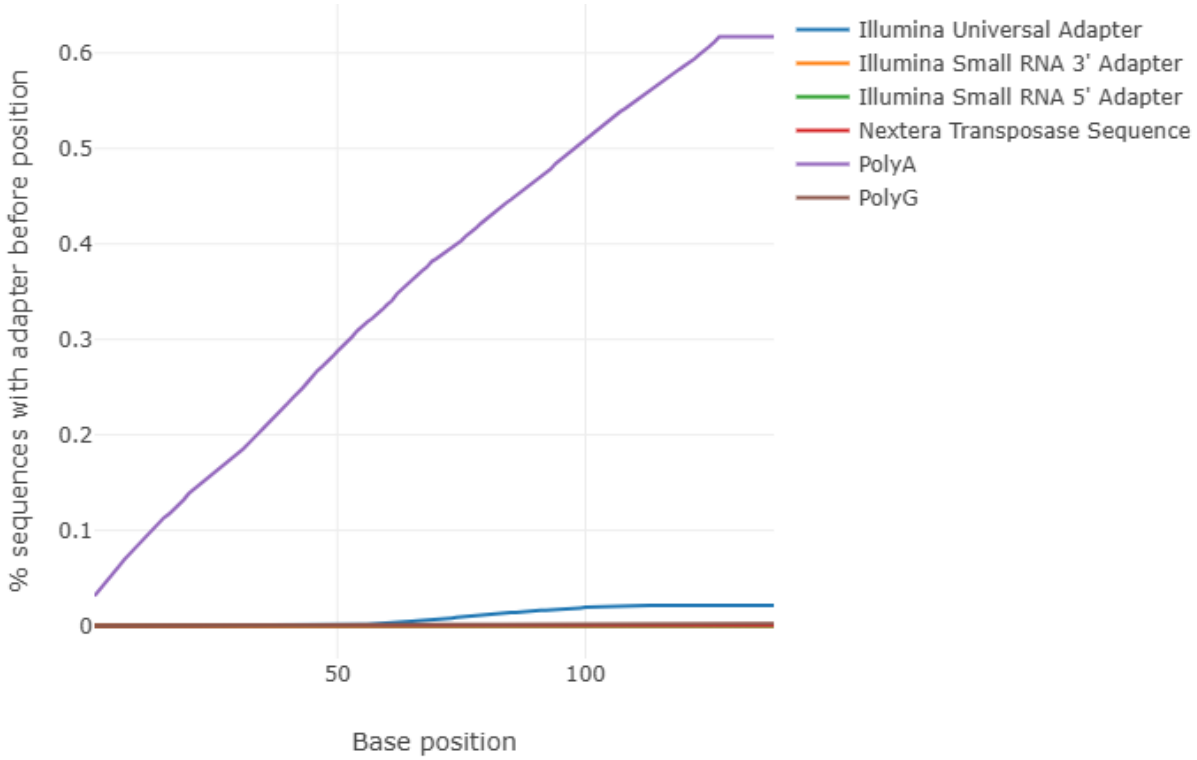
4.21 Sequence Duplication Levels : warn



4.22 Overrepresented sequences : pass

4.22.1 No overrepresented sequences

4.23 Adapter Content : pass

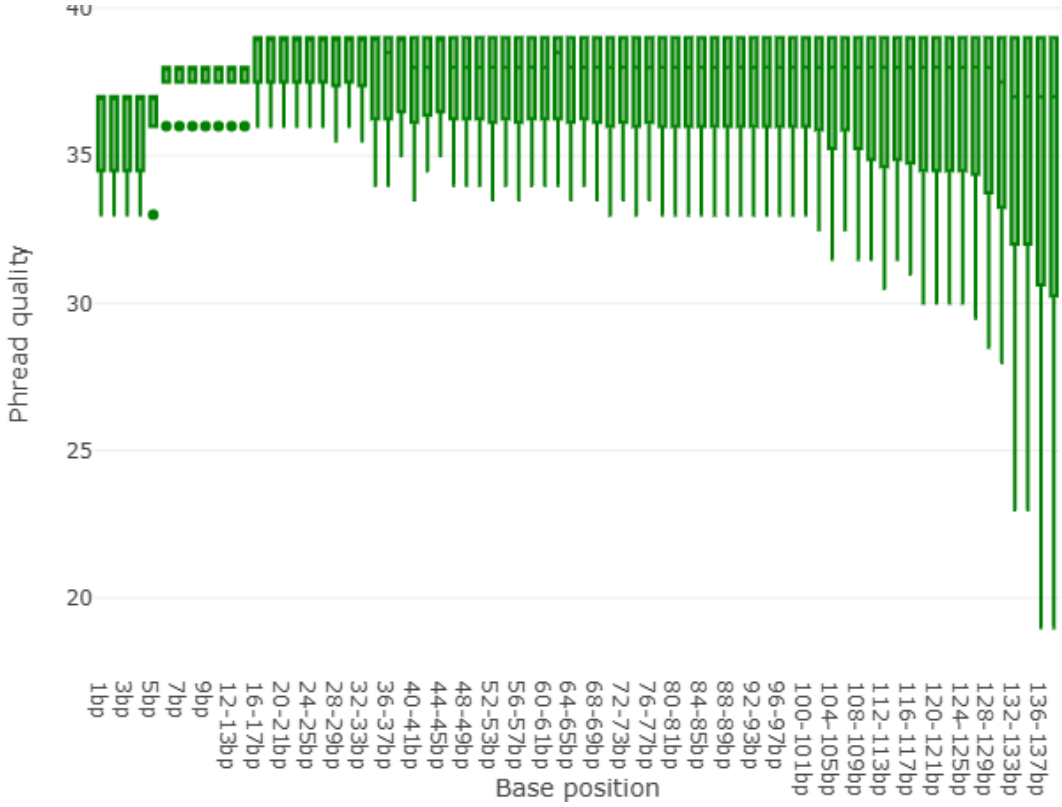


4.23.0.1 - Sample1_R2_fastq_gz_R2 paired

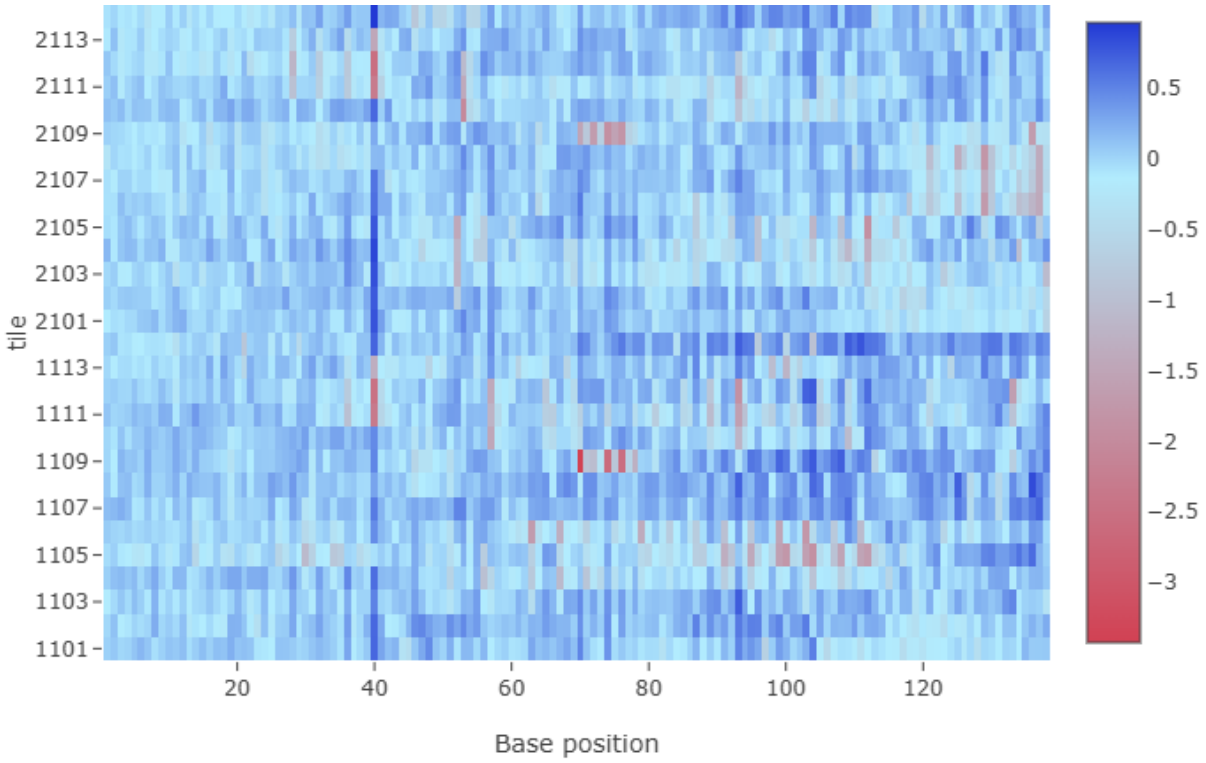
4.24 Basic Statistics: pass

Measure	Value
Filename	Trimmomatic on Sample1_0394321_L001_R2_001_fastq_gz _R2 paired_
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1206159
Sequences Flagged As Poor Quality	0
Sequence length	50 - 138
%GC:	42

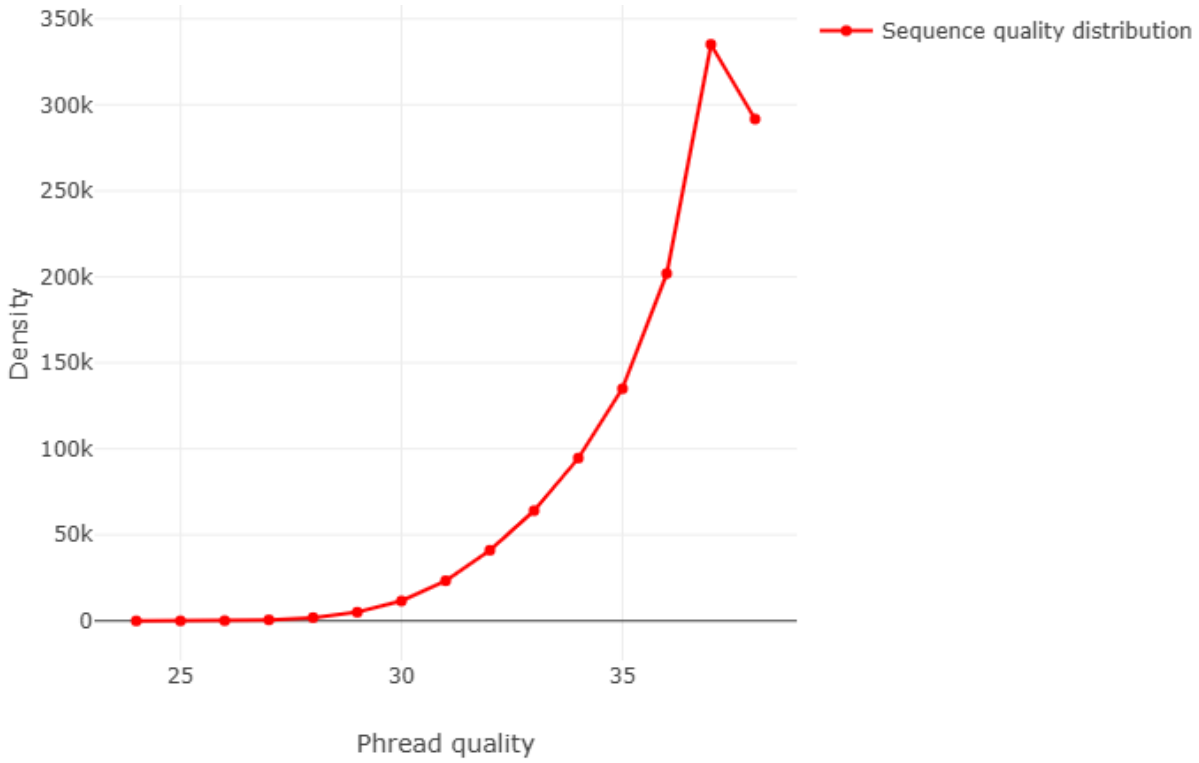
4.25 Per base sequence quality: pass



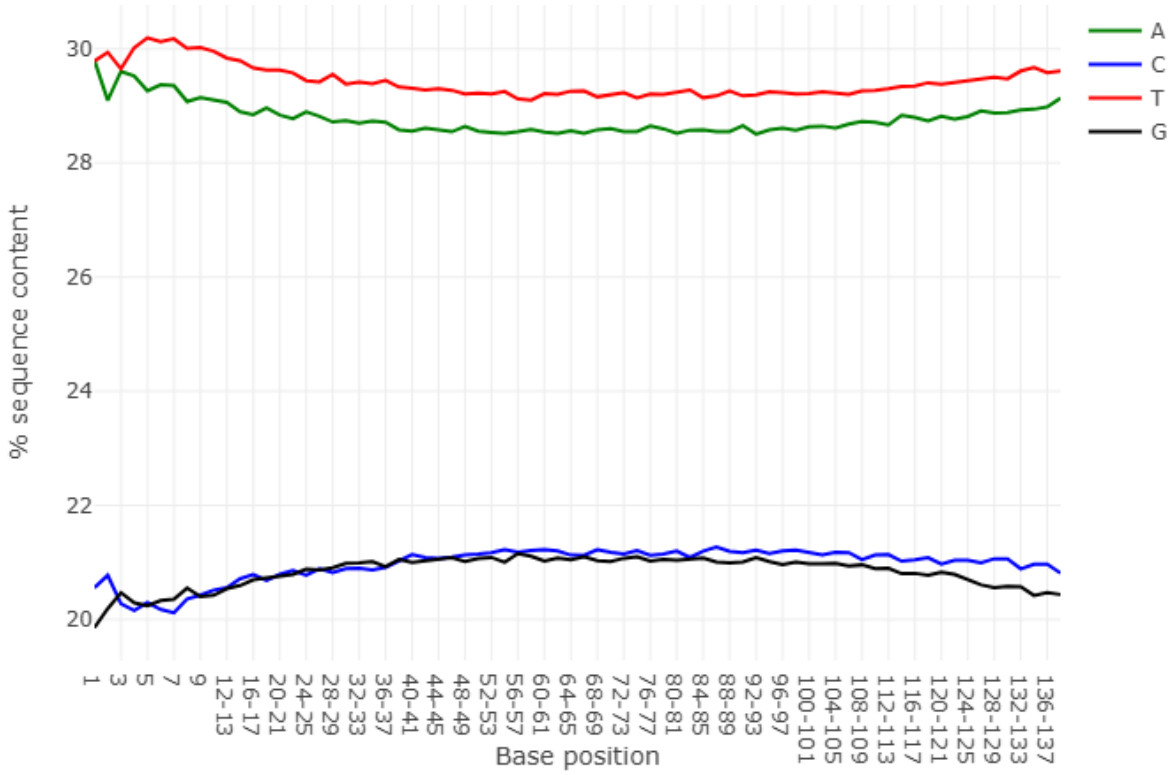
4.26 Per tile sequence quality: pass



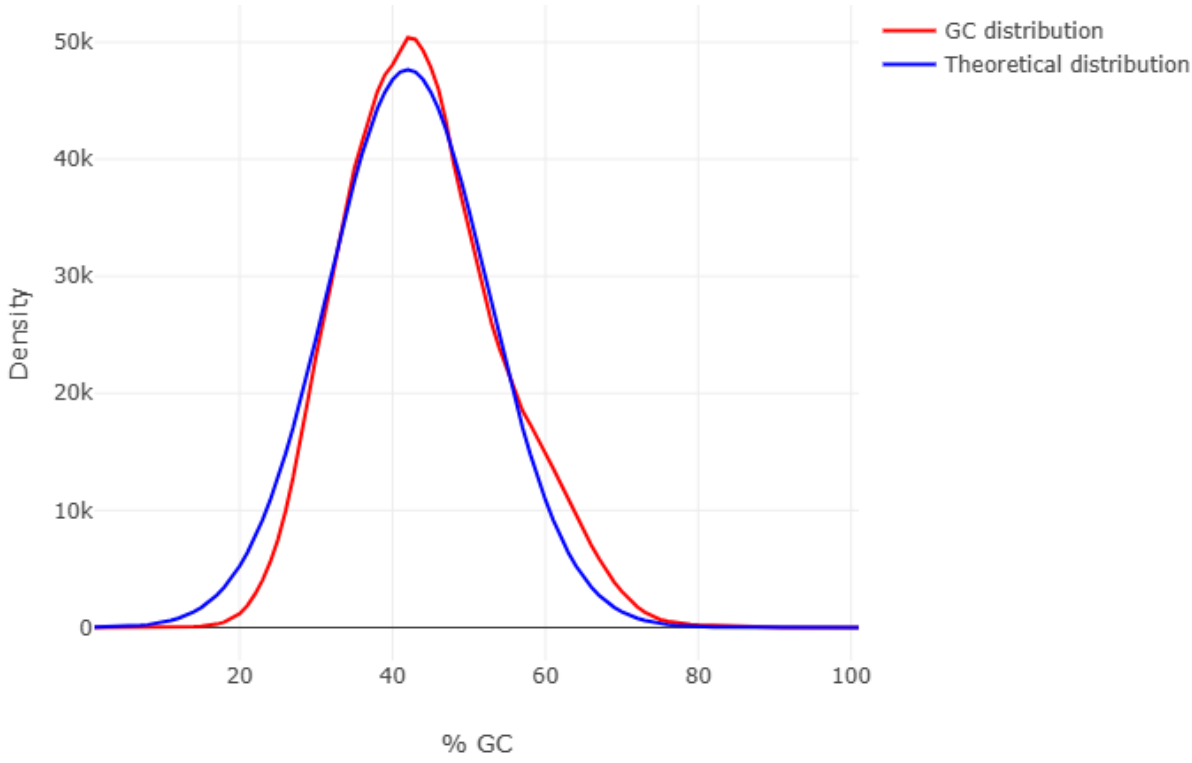
4.27 Per sequence quality scores: pass



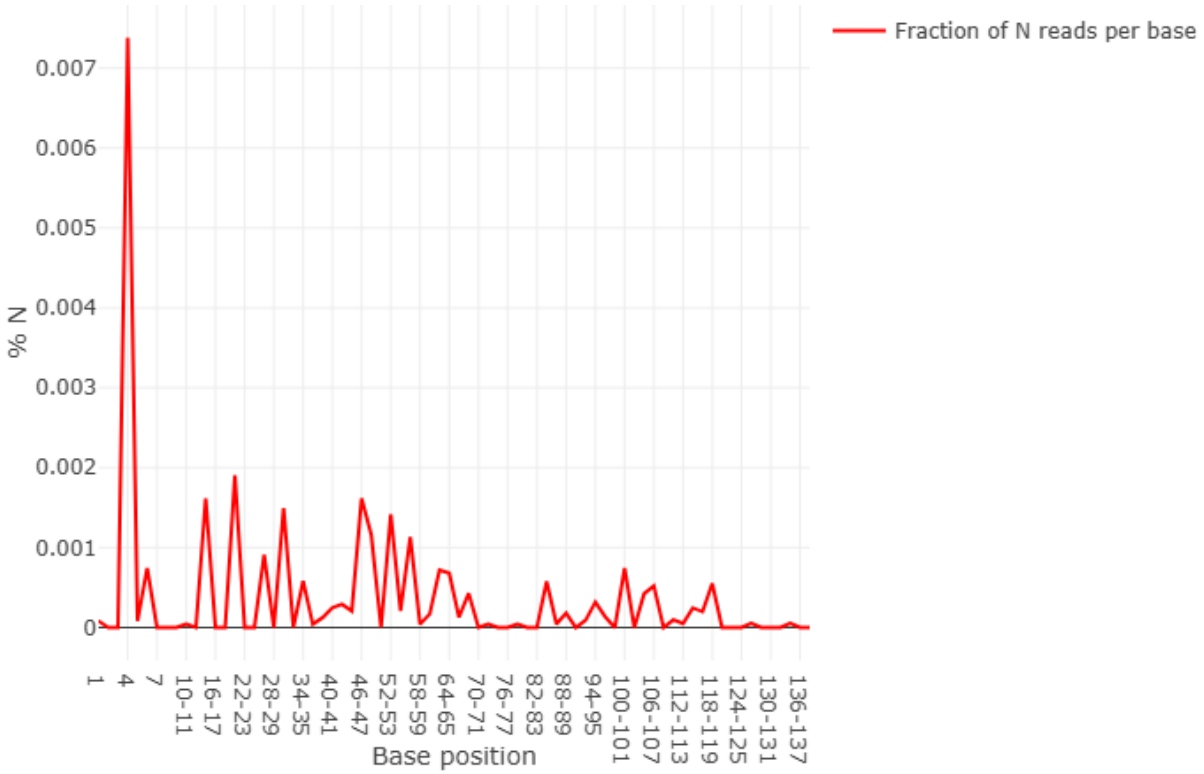
4.28 Per base sequence content : warn



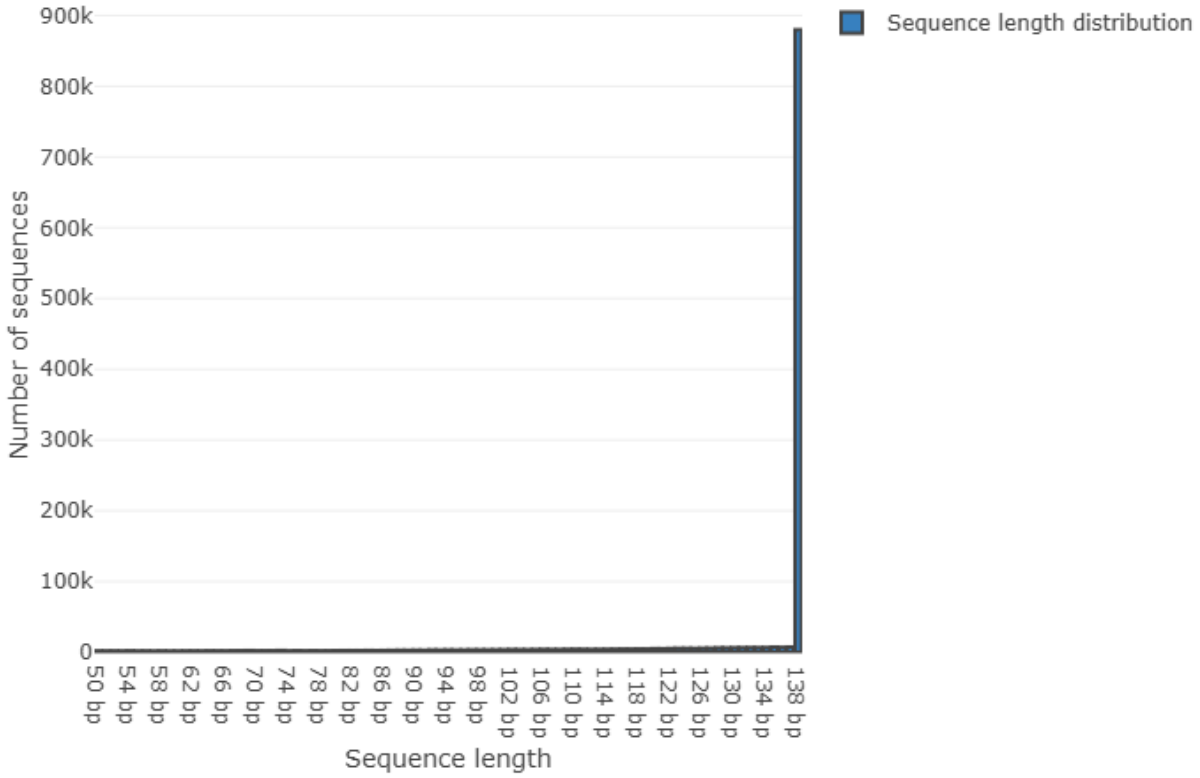
4.29 Per sequence GC content: pass



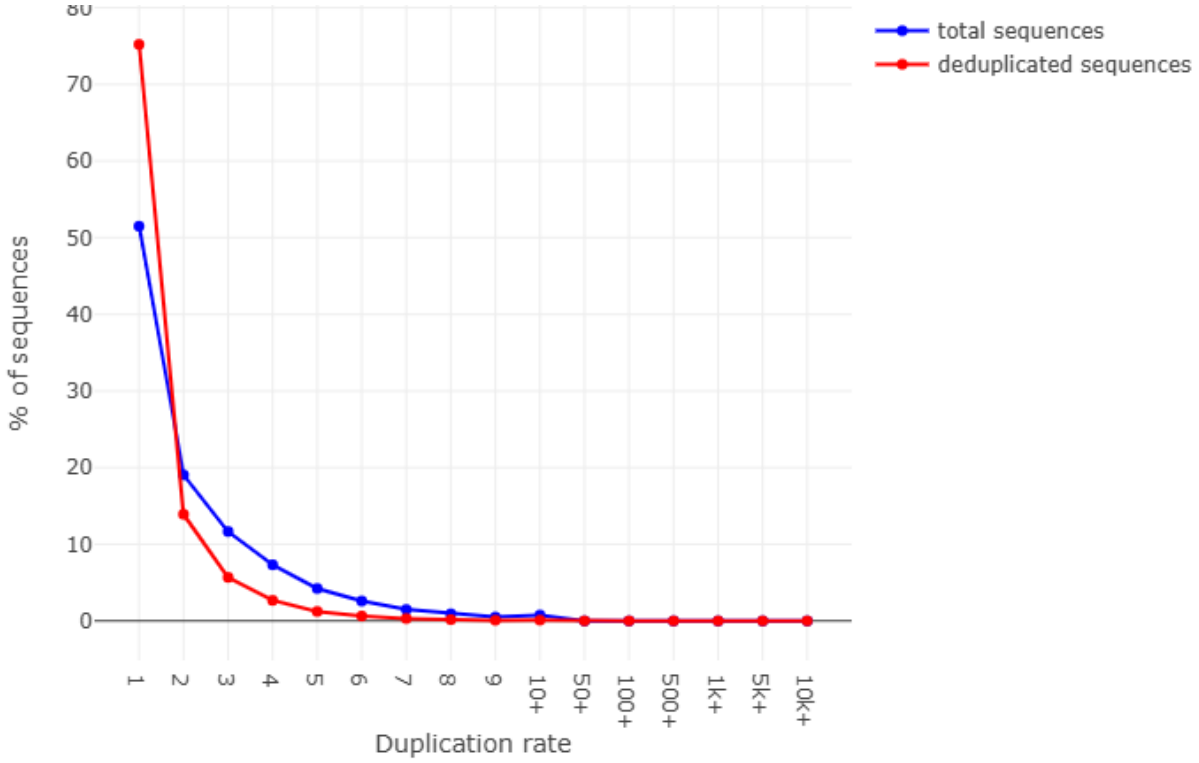
4.30 Per base N content : pass



4.31 Sequence Length Distribution : warn



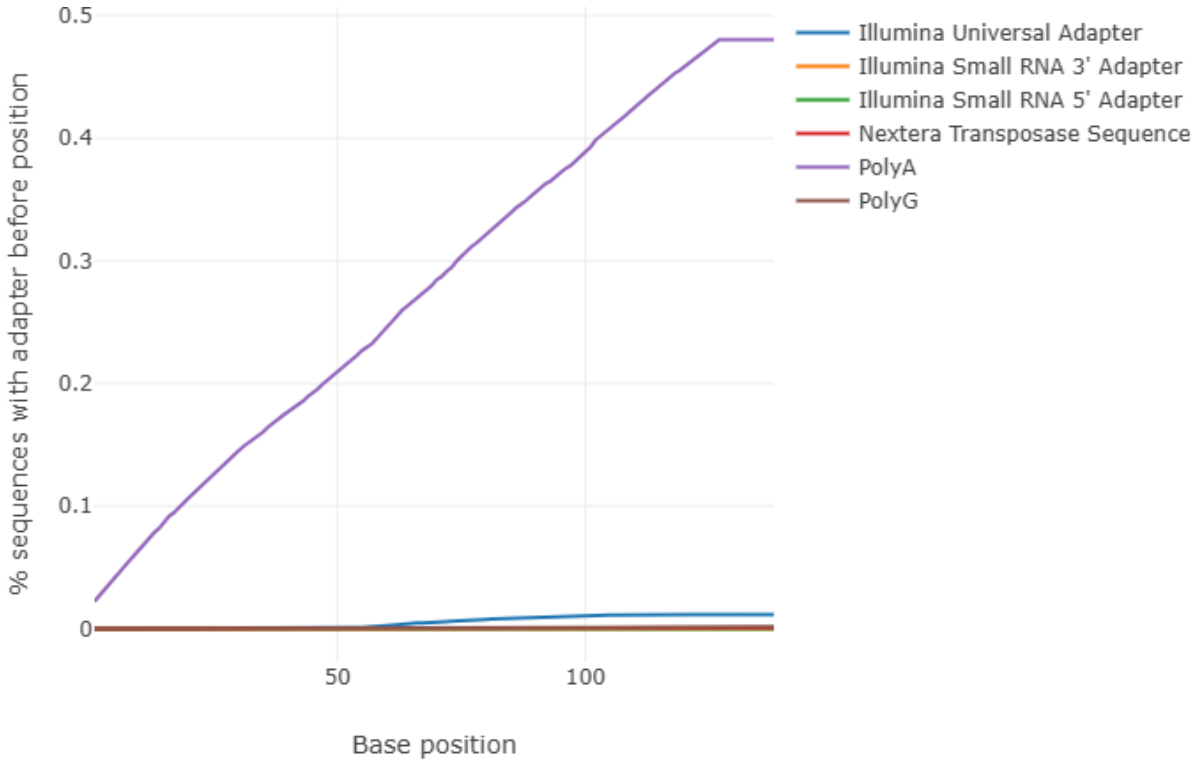
4.32 Sequence Duplication Levels : warn



4.33 Overrepresented sequences : pass

4.33.1 No overrepresented sequences

4.34 Adapter Content : pass



4.35 Read Alignment

Filtered reads were aligned to the appropriate human reference genome using a short-read aligner.

Reference genome:

Alignment tool:

4.35.1 Alignment Statistics

Key alignment metrics such as mapping rate and coverage were evaluated.

```
# Alignment statistics summary
```

4.36 Variant Calling

Variant calling was performed to identify single nucleotide variants (SNVs) and small insertions/deletions (indels) from the aligned reads.

Variant caller used:

4.36.1 Raw Variant Output

The initial variant call set contains all detected variants prior to filtering.

4.37 Variant Filtering

Variants were filtered based on quality metrics to reduce false positives. Filtering criteria included read depth, variant quality score, and allele frequency.

4.37.1 Filtering Criteria

- Minimum depth (DP):
- Minimum quality (QUAL):

```
# Summary of filtered variants
```

4.38 Variant Annotation

Filtered variants were annotated to identify affected genes and potential functional consequences.

Annotation tool/database:

5 Results

5.1 Variant Summary

A total of (X) variants were retained after filtering. These variants include SNPs and small indels distributed across multiple genomic regions.

```
# Tables or plots summarizing variants
```

5.2 Candidate Variants Related to Cardiomyopathy

Several variants were identified in genes previously associated with cardiomyopathy. These variants represent potential contributors to disease development.

6 Discussion

The identified variants provide insight into the genetic basis of cardiomyopathy in the analyzed patient sample. While some variants occur in known cardiomyopathy-associated genes, further validation is required to assess their pathogenicity.

6.0.1 Limitations

- Analysis based on a single sample
- Lack of functional validation
- Potential sequencing and alignment biases

7 Conclusion

This project demonstrates a reproducible workflow for variant discovery in cardiomyopathy using NGS data. The documented pipeline can be extended to additional samples and integrated with functional and clinical data for improved interpretation.

8 References

(To be added)